

Alessio Ludovico De Santis

Ph.D. Candidate in Astroparticle Physics

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Academic Interests

My research in **multi-messenger astrophysics** focuses on preparing for **next-generation observatories** such as **Einstein Telescope** and Cosmic Explorer. The core objective is to develop frameworks that combine more than a decade of gravitational-wave, electromagnetic, and neutrino data and models to maximize scientific return from future joint detections and constrain astrophysical source properties. My work primarily addresses two energetic transient classes: **core-collapse supernovae** and **binary neutron star mergers**. I study these phenomena through refining methods for parameter estimation, source localization, and identifying unique multi-messenger signatures to probe fundamental physics and extreme astrophysical environments, ultimately pushing the boundaries of multi-messenger astronomy for new insights into the universe's most violent events.

Education

- 2022–Present **Ph.D. in Astroparticle Physics**, *Gran Sasso Science Institute (GSSI)*
Advisors: Marica Branchesi, Samuele Ronchini, Filippo Santoliquido.
- 2022 **M.S. in Particle and Astroparticle**, *University "La Sapienza"*, (*Cum Laude*)
Advisors: Marco Vignati, Angelo Cruciani.
- 2020 **B.S. in Physics**, *University "La Sapienza"*
Advisors: Marco Vignati, Angelo Cruciani.

Academic Mentorship & Community Engagement

- Student **Master's Thesis Co-Advisor:** Guiding a master's student of Prof. Riccardo Murgia
Supervision on leveraging Dark Matter detectors for joint Supernova neutrino detection. Providing mentorship on how to write code to tackle this challenge.
- Peer **Outreach:** Speaker/Organizer for "Scienza in Bidda" (Sardinia, 2024), tutor for "Space
Mentorship Explorers" (GSSI, 2023–2026).
- and Outreach **GSSI Buddies Program:** Active mentor for incoming international students. I facilitate their integration into the institute and assist with Italian administration (banking, phones, bureaucracy) to ensure a smooth transition into life in Italy (2025).
GSSI Visitors Center Guide: Planning, preparing, and setting up interactive scientific experiments and demonstrations alongside conducting educational workshops and explaining the experiments to visitors (2026).

Visitings

- Oct 2025 **Visiting Ph.D. Researcher**, *Niels Bohr Institute*, Copenhagen, Denmark
Advisor: Prof. Irene Tamborra.
- Initiated a multimessenger project focused on Core-Collapse Supernovae.
 - Developed methodology to use neutrino signals as a template for generating Gravitational Wave (GW) waveforms.
- Apr–Sep 2024 **Visiting Ph.D. Researcher**, *Max Planck Institute for Physics*, Munich, Germany
Advisor: Dr. Federica Petricca (CRESST Group).
- Conducted simulations of TES-based cryogenic detectors to model unexpected sensor behavior called the Low Energy Excess.

Software Development contributions

- MAGGPY [\[GitHub\]](#) Public Monte Carlo code connecting BNS populations to sGRB observables. Leverages GW bounds to constrain jet properties and merger rates. (*under development*).
- GBM OSV [\[GitHub\]](#) Developed tools for background estimation in the Fermi-GBM detector. Used for the analysis of the "BOAT" (GRB 221009A) and ultra-long transients.
- CAPIBARA [\[GitHub\]](#) Simulation code for phonon propagation in cryogenic detectors. Improves predictive capabilities for CRESST experiment.
- GWFish Contributed to this open-source GW analysis code focused specifically on compact binary mergers. Implementing high-frequency corrections and arbitrary frequency or time series injection capabilities.

Talks & Conferences

- August 2026 **COSPAR Scientific Assembly**, *Contributed Talk*, Florence, Italy
Presented the MAGGPY code and its capabilities for constraining BNS population models
- May 2026 **SAlt Conference 2026**, *Contributed Talk*, L'Aquila, Italy
Presented my work and methodology for making joint predictions for next generation EM and GW detectors in "Constraining BNS Models with sGRB Data and Forecasting Next-Generation Synergies"
- Feb 2026 **THESEUS Conference 2026**, *Contributed Talk*, Malaga, Spain
Presented my results for calculating prompt detection prospects for the THESEUS mission in "E2E Forecasting of Multi-Messenger Synergies: Joint BNS-sGRB Detections with THESEUS, MAGGPY, and GWFish"
- Oct 2025 **Astro Seminar**, *Invited seminar*, Niels Bohr Institute, Copenhagen
"What gamma-ray burst observations tell us about the population of binary neutron star mergers?"
- 2025 - 2026 **ACME Tutorials**, *Tutorial Organizer*, GSSI
Organized multiple tutorials for the Astrophysics Center for Multimessenger studies in Europe (ACME), focused on the codes GWFISH and MAGGPY.
- 2025 **LVK Multi-Messenger Transients Group**, *Invited Presentation*, Online
"Constraining Binary Neutron Star Population Synthesis Models using Short Gamma-Ray Burst Data".
- 2025 **European Astronomical Society (EAS) Annual Meeting**, *Talk & Poster*
Presented work on binary star populations studies leveraging Fermi-GBM data.

- September 2024 **Young Researcher Meeting**, *Contributed Talk*, University of Palermo
"Using Dark Matter detectors as Neutrino Detectors".
- Dec 2024 **High Energy Colloquia**, *Invited Talk*, University of Cagliari
"CAPIBARA: phonon ray-tracing simulations for cryogenic dark matter detectors"
- July 2024 **CRESST Collaboration Meeting**, *Invited Talk*, LNGS
"Simulating the Double TES Response".
- Mar 2024 **High Energy Colloquia**, *Invited Talk*, University of Cagliari
"Cosmic Fireworks in Dark matter experiments - Supernova Neutrinos with COSINUS".
- Sep 2023 **109th SIF Congress**, *Contributed Talk*, Salerno, Italy
Talk on behalf of the RES-NOVA group (Award Winner).

Scholarships & Awards

- 2023 **"Premio per le migliori comunicazioni"**, *Società Italiana di Fisica (SIF)*
Awarded for the presentation at the 109th National Congress.
- 2022 **INFN Scholarship (N. 24318/2022)**
Awarded for master's thesis research contribution (3 k€).

Skills

- Languages **Italian and English:** Native/Bilingual proficiency. (Used daily in professional research, acquired through long-term residence in Canada and Italy). **German:** Professional working proficiency (Resided in Germany for 4 years, visiting researcher at Max Planck in Garching for 6 months). **Russian:** Proficient (Resided in Russia for 4 years). **Lithuanian:** Conversational (Resided in Lithuania for 4 years). **French:** Professional proficiency (Completed middle and high school curriculum in French).
- Programming **Python:** Advanced (Daily usage). Developer of public packages (MAGGPY, OSV) using Scipy, NumPy, emcee and dynesty for Bayesian inference. **C++:** Proficient. Utilized for hardware triggering systems and data analysis during B.S. and M.S. theses. **Tools:** Git (Version control for collaborative papers/code), $\text{\LaTeX}$ and Office productivity suite (Daily preparation of manuscripts and reports). **Additional:** Bash (Daily usage). Javascript for user interfaces, plugin development and API integration. HTML/CSS for applications and websites and SQL (tutored).

LIST OF PUBLICATIONS

Alessio Ludovico De Santis

Last Updated: July 2026

Refereed Journal Publications

- [1] **Constraining binary neutron star population synthesis models using short gamma-ray burst data**
A. L. De Santis, S. Ronchini, F. Santoliquido, and M. Branchesi
Astronomy & Astrophysics, Volume 710, A388 (2026).
DOI: [10.1051/0004-6361/202659597](https://doi.org/10.1051/0004-6361/202659597)

- [2] **Neutrino flux sensitivity to the next galactic core-collapse supernova in COSINUS**
G. Angloher, M. R. Bharadwaj, [...] A. L. De Santis, et al. (Corresponding author)
Journal of Cosmology and Astroparticle Physics (JCAP), Volume 2025, 037 (March 2025).
DOI: [10.1088/1475-7516/2025/03/037](https://doi.org/10.1088/1475-7516/2025/03/037)

- [3] **Enhanced athermal phonon responsivity in a kinetic inductance detector with integrated phonon collectors**
L. Pesce, A. L. De Santis, M. Calvo, M. Cappelli, U. Chowdhury, et al.
Applied Physics Letters, Volume 128, 163504 (2026).
DOI: [10.1063/5.0323811](https://doi.org/10.1063/5.0323811)

- [4] **Broadband Modelling of GRB 230812B Afterglow: Implications for VHE γ -ray Detection with IACTs**
S. Mohnani, B. Banerjee, D. Miceli, [...] A. L. De Santis, et al.
Astronomy & Astrophysics, Volume 705 (2026).
DOI: [10.1051/0004-6361/202556590](https://doi.org/10.1051/0004-6361/202556590)

- [5] **Crystal Eye: all sky MeV monitor with high precision real-time localization**
R. Aloisio, U. Atalay, B. Banerjee, [...] A. L. De Santis, et al.
Astroparticle Physics, Volume 174, 103171 (2025).
DOI: [10.1016/j.astropartphys.2025.103171](https://doi.org/10.1016/j.astropartphys.2025.103171)

- [6] **The Science of the Einstein Telescope**
A. Abac, [...] A. L. De Santis, et al. (Einstein Telescope Collaboration)
Accepted for publication in *Journal of Cosmology and Astroparticle Physics (JCAP)*.
[arXiv:2503.12263](https://arxiv.org/abs/2503.12263)

- [7] **Ultra-long MeV transient from a relativistic jet: a tidal disruption event candidate**
G. Oganessian, E. Kammoun, A. Ierardi, A. L. De Santis, et al.
Astronomy & Astrophysics (Letters), 703, L2 (2025).
DOI: [10.1051/0004-6361/202556591](https://doi.org/10.1051/0004-6361/202556591)

- [8] **Camelidae on BOAT: observation of a second spectral component in GRB 221009A**
B. Banerjee, S. Macera, A. L. De Santis, et al.

Astronomy & Astrophysics, 701, A68 (2025).
DOI: [10.1051/0004-6361/202554813](https://doi.org/10.1051/0004-6361/202554813)

Proceedings

- [1] **Preliminary tests of a phonon-mediated kinetic inductance detector with phonon-funneling volume.** L. Pesce, M. Calvo, M. Cappelli, U. Chowdhury, **A. L. De Santis**, et al.
A. L. De Santis
Il Nuovo Cimento C, 49, 21 (2026).
DOI: [10.1393/ncc/i2026-2602](https://doi.org/10.1393/ncc/i2026-2602)

- [2] **WINK: Advancing X and Gamma Ray Detection Technology for Space Observations.** T. Matteo, [...] **A. L. De Santis**, et al.
A. L. De Santis
PoS, ICRC2025, 857 (2025).
DOI: [10.22323/1.501.0857](https://doi.org/10.22323/1.501.0857)

- [3] **Supernova neutrinos at the RES-NOVA experiment**
A. L. De Santis
Il Nuovo Cimento C, 47, 303 (2024).
DOI: [10.1393/ncc/i2024-24303-9](https://doi.org/10.1393/ncc/i2024-24303-9)

Submitted Papers

- 4. **Gravitational wave detectability range informed by external messengers**
S. Ronchini, A. Chopra, T. Dal Canton, B. Banerjee, **A. L. De Santis**, M. Branchesi
Submitted (2026).

- 5. **Direct Measurement of the Accretion Disk Formed in Prompt Collapse Mergers with Future Gravitational-Wave Observatories**
A. Dhani, A. Camilletti, **A. L. De Santis**, A. Cozzumbo, D. Radice, et al.
Submitted to *Physical Review Letters* (2025).
[arXiv:2507.14071](https://arxiv.org/abs/2507.14071)

- 6. **Assessing the Impact of Instrumental Requirements on the Scientific Performance of the Einstein Telescope**
U. Dupletsa, F. Iacovelli, [...] **A. L. De Santis**, et al.
Submitted to *Journal of Cosmology and Astroparticle Physics (JCAP)* (2026).
[arXiv:2607.27311](https://arxiv.org/abs/2607.27311)

- 7. **Assessing the Impact of Instrumental Requirements on the Scientific Performance of the Einstein Telescope**
U. Dupletsa, F. Iacovelli, [...] **A. L. De Santis**, et al.
Submitted to *Journal of Cosmology and Astroparticle Physics (JCAP)* (2026).
[arXiv:2607.27311](https://arxiv.org/abs/2607.27311)

- 8. **The ultra-fast afterglow of GRB 260226A**
B. Banerjee, [...] **A. L. De Santis**, et al.

Submitted to *Nature Astronomy* (2026).
[arXiv:2607.26129](#)

Works in Preparation

9. **Optimizing pointing strategies for immediate optical counterparts of Galactic Supernovae**
I. Giudice, **A. L. De Santis**, et al.
In preparation.
10. **Predicting Socioeconomic indicators from satellite data**
A. L. De Santis, C. J. González Cuatianquis et al.
In preparation.